Engineering and Regulatory Transitions in IS 3043: 2026: Moving from Conventional Earth Pits to Equipotential Safety Networks.

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The third revision of the Indian Standard for earthing, **IS 3043: 2026** (*Earthing — Code of Practice*), introduces fundamental philosophical and technical updates compared to the 2018 revision. The changes reflect alignment with the **CEA (Measures Relating to Safety and Electric Supply) Regulations, 2023**, **National Electrical Code (NEC) 2023**, **IS 732: 2019**, and **IS/IEC 61936-1**.

The standard shifts from archaic practices (such as relying on standalone earth pit resistance and treating earth as a return conductor) to an engineering framework driven by **equipotential bonding, Earth Fault Loop Impedance (Z_s), touch voltage limits (U_{Tp}), and Automatic Disconnection of Supply (ADS)**.

1. Demystifying Misconceptions: Isolated / "Single Earthing" vs. Integrated Bonding

One of the most consequential additions in the 2026 edition is **Clause 6.3 (*Mis Concepts in Earthing and the Recommendation*)**, which explicitly outlaws long-standing industry malpractices:

- **Ban on Dedicated/Isolated Earth Pits (Clause 6.3.1 & Fig. 23):** Providing isolated or independent earth electrodes for specific equipment (such as separate, unbonded earth pits for transformer body, transformer neutral, DG body, DG neutral, UPS, lifts, electronics, or Solar PV) is explicitly identified as an outdated misconception and a severe safety hazard. In real-world soil conditions, stray currents, soil resistance gradients, and mutual coupling inevitably generate dangerous potential differences between these unconnected electrodes during lightning strikes or power-frequency earth faults, endangering life and destroying connected equipment (see Fig. 23 below).

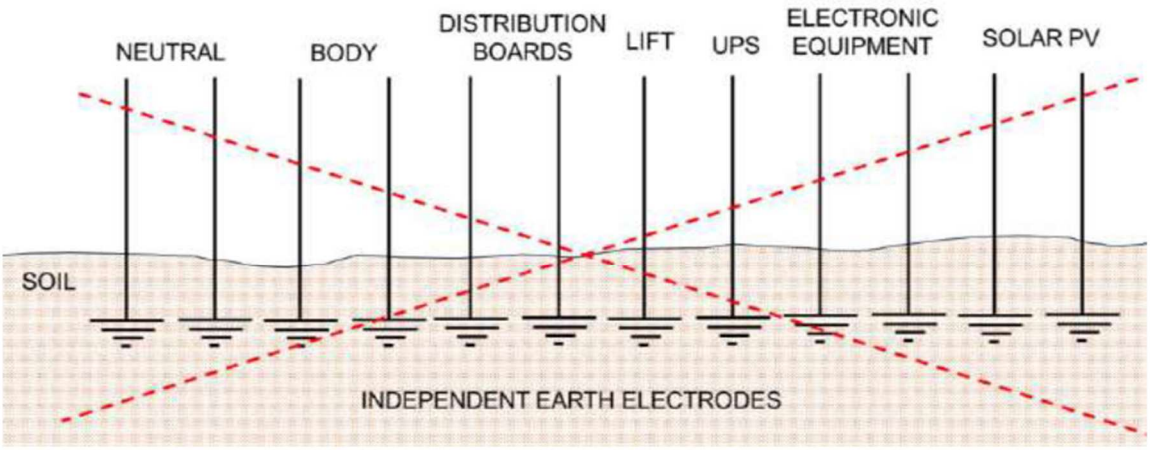


FIG. 23 MISCONCEPTION OF SEPARATE EARTH ELECTRODES

- **No Isolated "Clean" or "Instrument" Earths (Clauses 6.3.1.3, 6.3.1.4 & Fig. 24):** Creating segregated or unbonded earth electrodes for electronics, sensitive instrumentation, or Information and Communication Technology (ICT) in an attempt to attain a "noiseless" or

"clean" reference is strictly prohibited for general installations. Rather than eliminating noise, separating electronic, power, lightning (LPS), and renewable earthing networks compromises personnel safety at power frequencies, creates severe potential rise differences across interconnected equipment, and completely fails to provide high-frequency Electromagnetic Compatibility (EMC) (see Fig. 24 below).

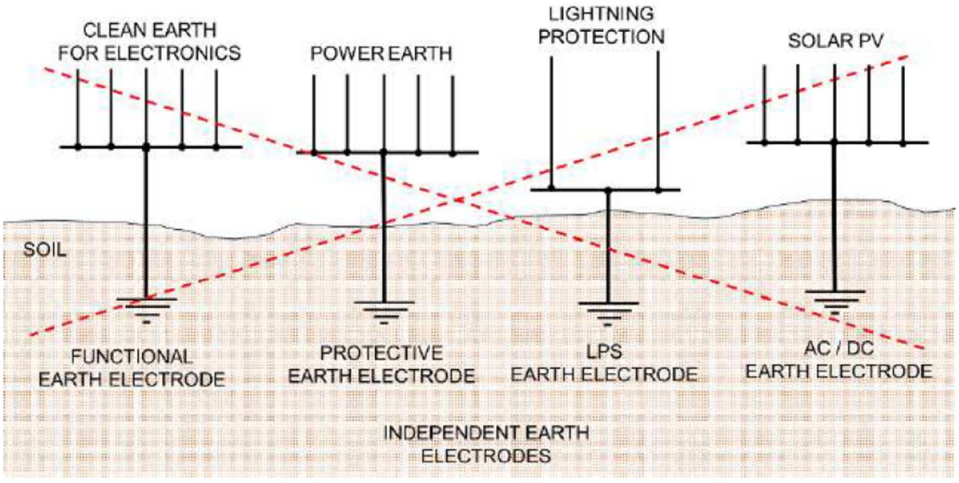


FIG. 24 MISCONCEPTION OF 'DEDICATED', 'INDEPENDENT', OR 'ISOLATED' EARTH ELECTRODES

- **The Single Integrated Earthing Network (Clauses 6.3.2, 7.2.1 & Figs. 26/27):** Rather than keeping systems separated or relying solely on buried interconnected loops, all functional earthing, protective earthing, lightning protection systems (LPS), and renewable energy terminations must interconnect into an equipotential bonding network tied directly to the Main Earthing Terminal (MET). Connecting all metallic services and installations to the MET creates a three-dimensional equipotential zone that eliminates dangerous potential gradients and guarantees electromagnetic compatibility (EMC) (see Fig. 26 / Fig. 27 below).

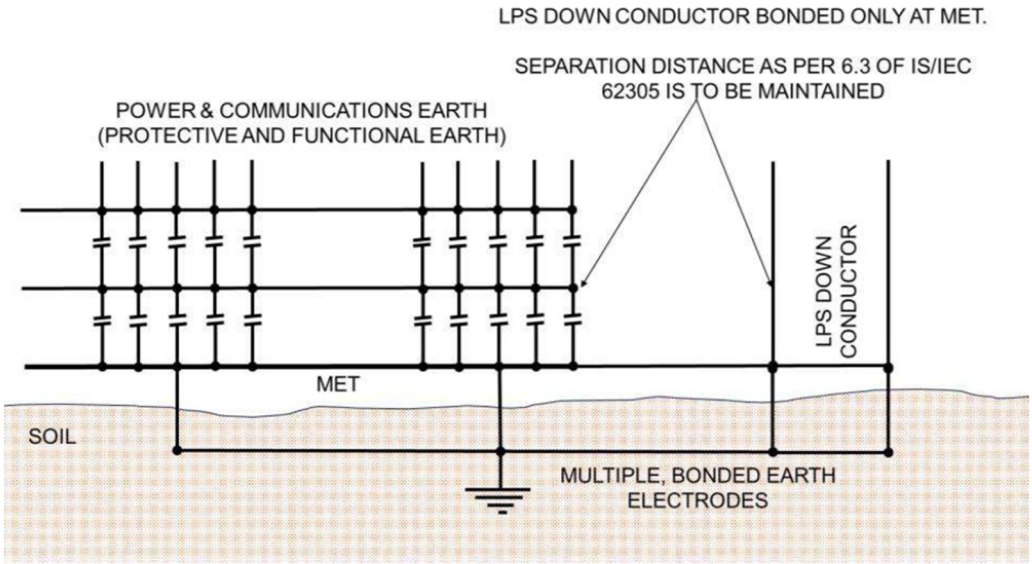


FIG. 26 RECOMMENDED CONFIGURATION FOR THE EARTH ELECTRODES, EARTHING NETWORK AND AN ELECTRICALLY INSULATED LPS

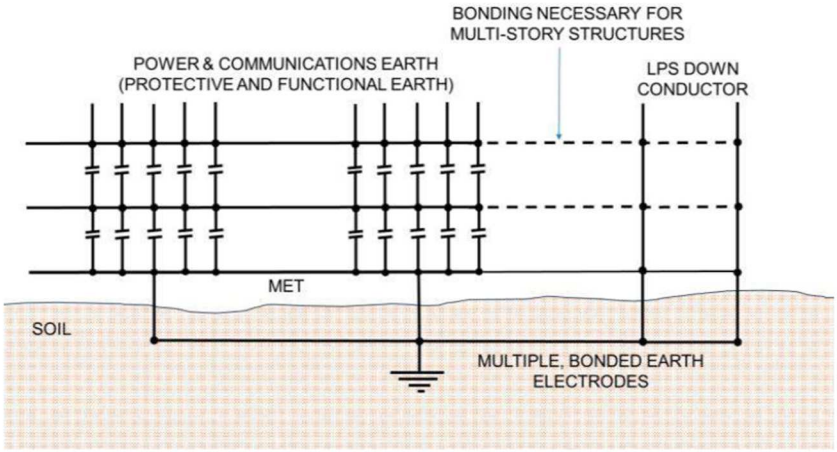


FIG. 27 RECOMMENDED CONFIGURATION FOR THE EARTH ELECTRODES, EARTHING NETWORK AND ATTACHED LPS

- **Single vs. Duplicate Earthing on Equipment (Clauses 4.2.4, 4.5.1 & Figs. 33A, 33B, 33C):** While conventional rules historically demanded two separate earth connections for all equipment, IS 3043: 2026 clarifies that the protective conductor connection may be restricted to a single connection if only one terminal is provided by the manufacturer in compliance with relevant product standards (such as IS/IEC 61439 for switchboards). Forcing an unapproved second terminal or hole into factory-tested equipment voids product certifications and risks mechanical damage. Furthermore, each individual protective or bonding conductor must connect securely to its own designated terminal on the busbar; stacking multiple conductors onto a single bolt provision is strictly prohibited (see Figs. 33A, 33B, and 33C below).

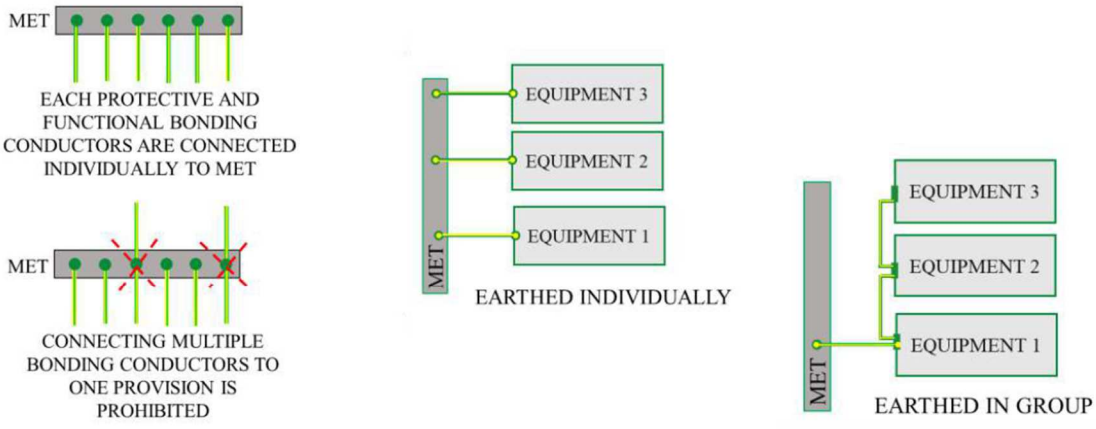


FIG. 33A FIG. 33B FIG. 33C

FIG. 33 (A, B AND C) EXAMPLE OF CONNECTING PROTECTIVE CONDUCTORS INDIVIDUALLY TO MET, EARTHING INDIVIDUALLY AND EARTHING IN GROUP

*"**Clause 4.2.4** All equipment of voltages of 250 V to 1 000 V shall be earthed by two separate and distinct connections with earthing arrangement. However, the number of connections shall be restricted to one, if only one provision is provided by the equipment manufacturer, and the equipment conform the relevant IS/IEC standard."*

***"Clause 4.5.1** Two distinct and separate earth connections are recommended (called as double earthing), to ensure redundancy in protective earthing. However modern, indoor/outdoor equipment cannot have provision for double earthing. In such case modifying the equipment for making additional provision for second protective earth connection may damage the equipment. Under such condition duplicate connection should not be required, provided the class 1 equipment satisfies the requirement of the relevant product standard."*

2. Global Earthing System (GES)

- **Definition & Concept (Clauses 3.33 & 10.2.4):** A Global Earthing System (GES) is an integrated, equivalent earthing network formed by the close physical proximity and interconnection of local earthing arrangements. This encompasses underground metallic cable sheaths, continuous protective multiple earthing (PEN) conductors, building foundation earth electrodes, and continuous metallic utility pipes **across an urban or industrial area**.
- **Elimination of Dangerous Touch Voltages (Clause 10.2.4):** In traditional isolated setups, an earth fault produces severe local Earth Potential Rise (EPR / U_E). In contrast, a GES acts as a broad quasi-equipotential surface across urban or industrial areas. By subdividing prospective earth fault currents among hundreds of parallel paths, it suppresses local ground potential gradients so effectively that touch voltages (U_T) are kept well within the permissible thresholds (U_{Tp}) specified in IS/IEC 60479-1.
- **Design Relief & Substation Optimization (Clauses 10.2.4, 10.3 & Figs. 57/63):** In an area verified as a GES (such as dense city centers, interconnected utility distribution grids, or expansive industrial complexes), step voltage and transferred potential measurements are not required. Furthermore, substation earth grids no longer need to be excessively over-dimensioned under the flawed assumption that 100% of the fault current enters the local earth pit; the metallic and parallel interconnected paths divert the bulk of the return current directly back to the source (see **Fig. 57** for current division and the GES design criteria in **Fig. 63** of IS 3043: 2026).
- **Engineering & Design Relief (Clauses 10.2.4 & 10.3):** For any installation verified to exist within a Global Earthing System (e.g., dense commercial city centers, contiguous distribution substation networks, or large industrial plants), the standard grants key engineering exemptions:
  - **No Separate Step Voltage or Transferred Potential Measurements:** Because the ground surface and interconnected equipment metalwork rise and fall in potential together, hazardous potential differences across a walking pace or hand-to-foot contact cannot physically develop.
  - **Optimized Substation Earthing Design:** Substation earth grids no longer need to be excessively over-dimensioned based on the conservative assumption that 100% of the fault current enters the local soil. Instead, calculations account for the fact that

the vast majority of fault current returns through metallic paths (such as cable screens, continuous overhead earth wires, and bonded structural conductors), allowing individual earth electrodes to be sized only for the minor fraction of current actually dissipated into the local earth.

3. Neutral Earthing & System Earthing Concepts

- **<u>System-Referencing-Conductor (SRC) (Clauses 3.65 & 6.2):</u>**
  - **Elimination of Ambiguous Terminology:** In earlier editions, the physical link connecting a power source neutral or star point to the earthing electrode was loosely termed an "earthing lead" or "neutral grounding conductor," causing frequent confusion with equipment protective earth (PE) conductors. The 2026 standard formally adopts the precise term **System-Referencing-Conductor (SRC)**.
  - **Redefining the Purpose of Earthing:** Modern power systems no longer treat the soil as an active return path for earth fault currents. The SRC's primary function is to **fix or anchor the potential of current-carrying live conductors** with respect to the general mass of the earth.
  - **Overvoltage Suppression & System Insulation:** By firmly tying down this reference potential, the SRC stabilizes phase-to-earth voltages, prevents dangerous neutral displacement during ground faults, restricts insulation stresses across connected equipment, and damps severe transient or resonant overvoltages.
- **Mandatory Utility PME (TN-C) vs. Legacy Consumer TT Networks (Clauses 4.2.1, 4.2.17.2 & 6.2.3):**
  - **The Statutory Utility Mandate (Clause 4.2.1):** For public low-voltage distribution up to 1 kV, IS 3043: 2026 mandates that the DISCOM implement a **TN-C system with Protective Multiple Earthing (PME)** in compliance with Central Electricity Authority (Safety) Regulations 16, 17, 18, 43, and 74. Under this requirement, the utility must combine the neutral and protective earth into a continuous **PEN conductor** and provide an official earthed terminal at the consumer's premises.
  - **Why TT Remains in the Standard:** The standard includes comprehensive rules and calculation methods for **TT systems** (Clauses 6.2.3.2 and 7.4.3) not to offer DISCOMs an alternative network design, but to address the reality of legacy overhead networks (such as in Kerala). Where the utility supplies only phase and neutral without an extended earthed terminal, the consumer is forced to rely solely on an independent local earth electrode, effectively operating as a TT installation.
  - **The Fault Loop Bottleneck in TT:** In a TT arrangement, earth fault return current must pass through high-resistance soil ($R_A + R_B$), choking the fault current so severely that standard overcurrent protective devices (MCBs/MCCBs) cannot operate. For this reason, Clause 4.2.17.2 mandates the use of an **RCD** for automatic disconnection within the required time limits on all TT installations.

- **Key Prerequisites for Converting Legacy TT to TN-C-S (PME) (Clauses 4.2.1, 4.2.2 & 4.2.16):**
  - **Strengthening the PEN Conductor Against Open-Circuit Faults:** The primary danger of TN-C-S is a snapped or open neutral line, which can elevate all bonded consumer metalwork to line potential (230 V). Converting legacy open-wire networks requires robust, high-tensile conductors—such as Aerial Bunched Cables (ABC)—complying with the minimum statutory sizing of 10 mm^2 **Copper or** 16 mm^2 **Aluminium**.
  - **Distributed Multi-Point Earthing:** The utility must earth the PEN conductor not only at the distribution transformer, but also at the extremities of every distribution spur and at intermediate points along the mains. This distributed grounding ensures that if a conductor break occurs, downstream equipment frames remain tied close to earth potential.
  - **Local Redundancy at the Consumer Intake:** Under IS 3043: 2026 (Clause 4.2.3 & Figs. 3A–3C), the consumer receiving a supply exceeding 250 V must also connect an **additional earth electrode** to the Main Earthing Terminal (MET) at the service intake, providing local parallel redundancy with the utility grid.
- **Intake Separation Link & Absolute Prohibition on Downstream Recombination (Clauses 4.2.3 & 4.2.16):**
  - **The Intake Test Link:** At the consumer's main distribution board or metering point, the incoming utility PEN conductor connects to the earthing arrangement and is split into two separate conductors: the **Neutral conductor (N)** and the **Protective Earth conductor (PE)** via a removable link (provided for mandatory insulation resistance testing).
  - **Permanent Separation in Branch Circuits:** Once divided at the intake, **PEN, PEM, or PEL conductors are strictly prohibited downstream**. Recombining the Neutral and Earth wires anywhere inside the premises is forbidden, as it forces continuous load return currents into equipment chassis, conduits, and cable trays—creating hazardous touch voltages, fire risks, high-frequency electromagnetic interference (EMI), and nuisance tripping of RCDs.
- **Standby and Parallel Generation Earthing (Clause 9):**
  - **Standby Generators (Clause 9.2.2.1.2):** Standby generating sets operating without mains paralleling facilities require **4-pole changeover switching** to isolate the generator neutral from the utility neutral earth. In an existing utility TN-C-S supply with certified PME, a 3-pole changeover with a linked neutral may be used.
  - **Parallel Generators & Third Harmonic Currents (Clause 9.2.2.2):** To prevent dangerous circulating third-harmonic neutral currents during parallel generator operation, IS 3043: 2026 permits three neutral grounding architectures: interconnecting all generator neutrals via a single neutral bus linked to the MET, installing interlocked alternator star-point switches (so only one machine neutral is earthed at any time), or providing a dedicated Neutral Earthing Transformer (NET) solidly tied to the main busbar.

4. Structural Earthing & Foundation Earth Electrodes

- **Structural Earth Electrodes Recognized (Clauses 3.62 & 8.2.5):**
  - Metal reinforcement bars (rebars), steel sheet pilings, and structural steel frameworks embedded in concrete or driven into the earth are formally classified as structural earth electrodes.
  - They serve as natural, highly effective earthing elements that fulfill all electrical safety requirements without compromising the structural integrity or mechanical design of the building.
- **Foundation Earth Electrodes (Clause 8.2.5):**
  - Recognized as the most reliable, permanent, and cost-effective electrode configuration for both 50 Hz power-frequency fault dissipation and lightning protection system (LPS) discharge.
  - Because it spans the entire building footprint, it provides the lowest achievable earth impedance, remains unaffected by seasonal surface droughts or topsoil freezing, and establishes an optimal equipotential plane for the entire installation.
- **Behavior of Steel Fully Encapsulated in Concrete (Clause 8.2.5.1):**
  - When bare steel or galvanized steel is completely embedded within foundation concrete at a depth of at least 5 cm, the high alkalinity of the concrete mix ($pH > 12$) passivates the metal surface.
  - As long as it remains entirely encapsulated and protected from soil contact, the steel is naturally safeguarded against oxidation and corrosion for the operational lifespan of the structure.
  - (Regional Caution: In coastal regions or environments with saline groundwater, zinc coatings on galvanized steel can react aggressively with penetrating chlorides and damage the concrete; hence, bare rebar or stainless steel is preferred in such conditions).
- **Galvanic Coupling Hazard at the Concrete-to-Soil Transition (Clauses 8.1.5.3 & 8.2.5.4):**
  - Steel embedded in concrete assumes a noble electrochemical potential practically equivalent to **copper in soil**.
  - If a steel conductor extends straight out from inside the foundation concrete directly into the adjacent damp soil, a strong **electrochemical cell** is formed with an active driving potential of approximately 1 V.
  - Under this galvanic couple, the extensive steel rebar grid inside the concrete becomes the cathode, while the small steel conductor in the soil acts as the sacrificial anode,

leading to rapid dissolution and mechanical failure of the conductor right at the soil interface.

- **Mandatory Material & Installation Rules (Clauses 8.1.5.3 & 8.2.5.3):**
  - **Prohibition on Bare/Galvanized Steel in Soil Contact:** Any earth electrode or bonding conductor that transitions directly from the foundation concrete into open soil **must not be bare or galvanized steel** (post-applied paint or standard zinc galvanizing is strictly non-compliant for this application).
  - **Permitted Transition Materials:** Only **stainless steel, copper, or copper-bonded steel** conductors are permitted where conductors pass from the concrete directly into surrounding earth.
  - **Above-Ground Terminal Routing:** Connection terminal lugs exiting the concrete foundation to link with the Main Earthing Terminal (MET) or testing chambers should emerge **above ground level** (either inside the electrical room or along external walls) to ensure they never come into direct contact with soil moisture.

5. Other Critical Upgrades to Note

- ✓ **Protective Earthing (PE) vs. Functional Earthing (FE) Integration (Clauses 7.2.1, 7.6.8 & 7.7):** IS 3043: 2026 resolves long-standing industry confusion regarding "clean/instrumentation earthing." While safety PE always takes precedence, all functional earths (FE) and functional bonding conductors (FB) must terminate at a **Main Functional Earthing Terminal (MFET)**, which is then **bonded to the Main Earthing Terminal (MET) only once**. Independent functional earth pits in soil are prohibited. To prevent high-frequency noise loops, FE conductors must be insulated throughout their run, dimensioned to a minimum of 2.5 mm^2 Cu (with protection) or 4 mm^2 Cu (without protection), and identified by the alphanumeric notation **FE** or colored **Cream**. Using the bi-color **Green-and-Yellow for functional earthing conductors is strictly prohibited**.
- ✓ **Touch & Step Voltage Criteria (Clause 5.2):** Replaces simplified empirical assumptions with body impedance curves based on **IS/IEC 60479-1: 2018** (5% ventricular fibrillation threshold curve c_2). The revised permissible touch voltage (U_{Tp}) is dynamically tied to fault duration.
- ✓ **Temporary Overvoltages (TOV) & HV/LV Earthing Interaction (Clause 5.3):** Clear criteria are established for interconnecting or separating HV (R_E) and LV (R_B) substation earths based on power-frequency stress voltages (U_1, U_2) and fault voltages (U_f) to safeguard LV installations from HV side earth faults.
- ✓ **Maximum Disconnection Times (Table 1):** Strict adherence to disconnection times for low voltage circuits (e.g., 0.4 s for final circuits ≤ 63 A socket outlets / 32 A fixed equipment in 230 V TN systems, and 0.2 s for TT systems).
- ✓ **RCD Selection & dc Fault Considerations (Clause 7.5.7):** Details application of **Type AC, Type A, Type F, and Type B RCDs**. Standard Type AC RCDs are banned where electronic power

supplies/LEDs are present. Type B RCDs are explicitly mandated for three-phase drives, EV charging infrastructure, and transformerless Solar PV installations.

- ✓ **Earth Enhancing Compounds Curtailed (Clause 8.1.7):** Usage of chemical backfill / enhancing compounds is explicitly **discouraged** and limited strictly to rocky/difficult terrains where soil resistivity exceeds $3\,000\ \Omega \cdot \text{m}$ (and must conform to IS/IEC 62561-7 / IS 18925 Part 7). They are prohibited in soils below $200\ \Omega \cdot \text{m}$ due to accelerated conductor corrosion.

Conclusion

The release of IS 3043: 2026 marks a decisive shift away from empirical, pit-centric earthing towards a unified, scientifically grounded safety architecture. By prioritizing protective equipotential bonding, enforcing strict Earth Fault Loop Impedance limits, and debunking dangerous misconceptions around isolated "dedicated" earth pits, the standard addresses modern electrical environments characterized by power electronics, distributed solar generation, and structural complexity. For distribution utilities and practicing engineers alike, the transition to statutory PME (TN-C-S) and integrated foundation grounding is no longer just a regulatory compliance matter—it is the prerequisite for reliable automatic disconnection, electromagnetic compatibility, and true personnel safety.